TWO OUT OF THREE COMPUTING A RIGHT DERIVED FUNCTOR FORCES THE THIRD

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5241-5254, immediately following the statement at 5233-5239.

Editorial note. The LF case is dual and carried by the same argument.

Allowed standard prerequisites (editorial).

- The 2-out-of-3 property for definedness. Lemma lemma-2-out-of-3-defined of this chapter, cited at proof lines 5241-5244. RF is defined at Z and RF applied to the triangle produces a distinguished triangle.
- The situation. Situation situation-derived-functor of this chapter, cited at statement line 5234. Fixes F and the localization underlying RF.

Source reference. Original lemma 05SC: sources/stacks-derived/pinned-original-source.tex, source lines 4896–4896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05S8: sources/stacks-derived/pinned-original-source.tex, source lines 4736–4736. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 05S8. Let (X, Y, Z, f, g, h) be a distinguished triangle of $\mathcal{D}$. If X, Y compute RF then so does Z. Similar for LF.*

Proof. By Lemma 05SC we know that RF is defined at Z and that RF applied to the triangle produces a distinguished triangle. Consider the morphism of distinguished triangles

$$\begin{array}{c} (F(X), F(Y), F(Z), F(f), F(g), F(h)) \\ \downarrow \\ (RF(X), RF(Y), RF(Z), RF(f), RF(g), RF(h)) \end{array}$$

Two out of three maps are isomorphisms, hence so is the third. □

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